

HAIQING XU hxu489@gatech.edu

RESEARCH INTEREST

Human-Computer Interaction (HCI), Ubiquitous Computing, Computational toolkits, Behavior Change

EDUCATION

Georgia Institute of Technology, Atlanta, USA.

Ph.D. in Human-Centered Computing, School of Interactive Computing, Expected June 2028.

Advisor: Prof. Alex T Adam and Hyunjoo Oh

Selected courses: Designing Interactive Systems, User-Centered Design and Evaluation, Ubiquitous Computing, Cognitive Science.

Tsinghua University, Beijing, China.

Master of Fine Arts, Interdisciplinary Program of Art & Science, Jun 2023.

Advisor: Prof. Haipeng Mi

Selected courses: Human-Computer Interaction Technology, Interactive Media Design, Game Design

Beijing Institute of Fashion Technology, Beijing, China

Bachelor of Fine Arts, Product Design, Jun 2016

Selected courses: Prototyping and Product Development, Visual Communication, Design Thinking and Innovation

Polytechnic University of Milan , Milan, Italy

One-year exchange program. Oct 2014 - July 2015

Selected courses: Cross-Cultural Design Approaches, Design Theories and Practice

SELECTED PUBLICATIONS

Conference Publications

1. Haiqing Xu, Xiwen Yao, Sixuan Wu, Jung Hyun Bae, Zhifan Guo, Dian Lv, Zhihao Yao, HyunJoo Oh, Alex Adams. 2026. LumiBite: An In-the-Wild Technology Probe Exploring Personalized bottom-up Lighting Lunchbox for Enhanced Dining Experiences. In Proceedings of the Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '26).
2. Han Huang, Tingyu Cheng, Jinzhi Ye, Yuhui Zhao, Yiyang Wang, Haiqing Xu, Abigale Stangl, and HyunJoo Oh. 2025. Eliciting Design Guidelines of Paper-Based Tactile Interfaces for Eyes-Free Scenarios. In Proceedings of the Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '25).
3. Qiuyu Lu, Haiqing Xu, Yijie Guo, Joey Yu Wang, and Lining Yao. 2023. Fluidic Computation Kit: Towards Electronic-free Shape-changing Interfaces. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23).
4. Tianyu Yu, Weiye Xu, Haiqing Xu, Guanhong Liu, Chang Liu, Guanyun Wang, and Haipeng Mi. Thermotion: Design and Fabrication of Thermofluidic Composites for Animation Effects on Object Surfaces. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)

5. Guanhong Liu, Xianghua Ding, Jinghe Cai, Weiyun Wang, Xinyue Wang, Yuting Diao, Jin Chen, Tianyu Yu, Haiqing Xu, and Haipeng Mi. "Digital making for inheritance and enlivening intangible cultural heritage: A case of hairy monkey handicrafts." In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23).
6. Guanhong Liu*, Haiqing Xu*, Xianghua (Sharon) Ding, Mingyue Gao, Bowen Li, Fushen Ruan and Haipeng Mi. "It Puts Life into My Creations": Understanding Fluid Fiber as a Media for Expressive Display. (*Equal Contribution) (CHI '21).
7. Guanhong Liu*, Tianyu Yu*, Chun Yu, Haiqing Xu, Shuchang Xu, Ciyuan Yang, Feng Wang, Haipeng Mi, and Yuanchun Shi. Tactile Compass: Enabling Visually Impaired People to Follow a Path with Continuous Directional Feedback (CHI '21).

Journal Publications

1. Patel, Dinesh K., Ke Zhong, Haiqing Xu, Mohammad F. Islam, and Lining Yao. "Sustainable morphing matter: design and engineering practices." Advanced Materials Technologies 8, no. 23 (2023): 2300678.
2. Guanhong Liu, Tianyu Yu, Zhihao Yao, Haiqing Xu, Yunyi Zhang, Xuhai Xu, Xiaomeng Xu, Mingyue Gao, Qirui Sun, Tingliang Zhang, and Haipeng Mi. ViviPaint : Creating Dynamic Painting with a ThermoChromic Toolkit. Multimodal Technol. Interact. 2022, 6(8), 63, Paper

TEACHING EXPERIENCE

Semester Courses

1. Ubiquitous Computing, Graduate TA for CS 7470, Summer 2024, Fall 2024, Georgia Tech
2. Prototyping Interactive Systems, TA, Fall 2022, Tsinghua University

Workshops

1. Paper Interfaces Workshop: Usability and Design for Eyes-Free Scenarios, Georgia Tech, GA, October 2023.
2. Makers with traditional Hairy Monkey craft in Beijing, Tsinghua University, August 2022.
3. Fluid Fiber with Computing Systems, Tsinghua University, August 2021.
4. ViviPaint Workshop: Accessible ThermoChromic Painting Creation for Artists, Tsinghua University, July 2020.

RESEARCH EXPERIENCE

Uncommon Sense Lab | Georgia Institute of Technology | Research Assistant (Advisor: Prof. Alex Adam)

- Led the research project "LumiBite" that explores how ambient lighting influences food perception, a multisensory experience often studied in controlled lab settings with limited real-world applicability.

Codecraft Lab | Georgia Institute of Technology | Research Assistant (Advisor: Prof. Hyunjo Oh)

- Contributed to the paper titled 'Eliciting Design Guidelines of Paper-Based Tactile Interfaces for Eyes-Free Scenarios'.
- Co-hosted a participatory workshop with design students to gather insights and refine design guidelines.

Morphing Matter Lab, Carnegie Mellon University | Visiting Research Assistant (Advisor: Prof. Lining Yao)

- Co-Led and contributed to the paper titled 'Fluidic Computation Kit' (CHI'23)

The Future Laboratory, Tsinghua University | Master student (Advisor: Prof. Haipeng Mi)

- Contributed to collaborative projects: 'ViviPaint': a thermochromic toolkit for creating a dynamic painting.
- Worked on project of Thermotion (CHI '23), which proposed a novel design and fabrication method of using thermofluidic composites to create animation effects on two and three-dimensional surfaces and everyday material surfaces

Pervasive Human-Computer Interaction Laboratory | Research Assistant (Advisor: Prof. Chun Yu and Prof. Yuanchun Shi)

- Worked on the projects Tactile Compass (CHI '21), which focused on two novel tangible interfaces for navigation tasks for visual-impaired people.

AWARDS & HONORS

- 2022 The First Prize Scholarship, Tsinghua University
- 2021 The First Prize Scholarship, Tsinghua University
- 2016 Outstanding Graduate of Beijing (top 1%)

GRANTS & FELLOWSHIPS

Travel Grants

- Scholarship Recipient, CMD-IT/ACM Richard Tapia Celebration of Diversity in Computing (Sept 2024)
Selected for a competitive scholarship to attend the premier conference promoting diversity in computing.

INDUSTRY EXPERIENCE

Beijing Kuaipao Informational Technology Co., Ltd. | Product Manager | Beijing, China | Jul 2016 – Jul 2020

- 4 years of experience in product management, product marketing, business development
- Responsible for understanding guest and hotel needs, and identify and defining product requirements.

EXHIBITIONS

- Remaining Candle - New Media Art Installation | Beijing, China | Sep 2021
- China Public Art and Design Exhibition | Shenzhen, China | May 2021
- Qingdao Institute of Art and Science Innovation Inauguration Exhibition | Beijing, China | Oct 2021
- Zijin Award - 2nd National Student Design Exhibition | Nanjing, China | May 2021
- Tsinghua University Art Museum | Beijing, China | Jun 2020
- International Sleep Science and Technology Association (ISSTA) Berlin Headquarter | Berlin, Germany | Jun 2016
- Milan Furniture Fair | Milan, Italy | Apr 2015

SKILLS & LANGUAGES

Core Competencies

- Interaction Design: User-centered design methods, creative thinking across domains and technologies.
- Design Research Methods: Workshop facilitation, design probe, Wizard-of-Oz prototyping, participatory design, and cultural probes.
- Visual Communication: Graphics, videos, storyboarding, and wireframing.
- Problem-Solving: Self-directed, solution-oriented approach to challenges.
- Research Integrity: Commitment to transparency, ethics, and rigorous documentation.

Research

- Qualitative and Quantitative Research, Survey Design, Data Analysis (SPSS, SAS, MaxQDA).
- Usability testing, user interviews and surveys, pilot studies and large-scale studies, lab, field and remote studies.
- Quantitative usability tests (time spent doing the task, success, etc.) as well as physiological analysis.
- Design Methods: Workshop facilitation, design probe, Wizard-of-Oz prototyping, participatory design, and cultural probes.

Software& Hardware

- Software Developer,Python, C++, Arduino
- Hardware: Circuit design, embedded development
- Sensor & Data Integration: Unity/Android to BLE & WIFI EE modules/wearables
- User Interface (UI) : web-based prototyping (e.g., Figma, Adobe XD). Sketch.
- User Experience (UX), Sketches and mockup videos, design thinking, paper prototyping
- Communication: Blender, Premiere, Final Cut
- Fabrication & Prototyping: 3D/2D Modeling and Printing, Arduino, Laser cutting, CNC milling, Storyboards, Wireframes, Sketches, Animations, papermaking, Sewing, Knitting, Woodwork

Languages

- English (fluent), Chinese(native), Italian (beginner)